

Jay DeYoung

Allen Institute for AI
Seattle, WA USA
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I am a Postdoc (Young Investigator) at the Allen Institute for Artificial Intelligence. My main research interests are how to better use *all* of our data, be it to understand medicine today, or to drive scientific advancement for tomorrow.

EXPERIENCE

Allen Institute for Artificial Intelligence

Young Investigator

– Advisor: Doug Downey

Seattle, WA

Jan 2025 – Present

Northeastern University

Graduate Research Assistant

– Advisor: Byron C. Wallace

Boston, MA

Sept 2018 – Dec 2024

Amazon Comprehend Medical

Research Intern

Seattle, WA

Summer 2021

Allen Institute for Artificial Intelligence

Research Intern

Seattle, WA

Summer 2020

Raytheon BBN Technologies

Staff Scientist I & II

Cambridge, MA

2015-2018

Johns Hopkins University Human Language Technology Center of Excellence

Adjunct Research Associate

Baltimore, MD

2013-2014

EDUCATION

Northeastern University

Ph.D. in Computer Science, advised by Byron Wallace

– Thesis: “Automation Assistance For Systematic Reviewers”

2018 — 2024

Johns Hopkins University

M.S.E. in Computer Science

2014

Johns Hopkins University

Bachelor's in Computer Science and Applied Mathematics

2014

SERVICE

I regularly review for *ACL conferences: ACL, EMNLP, NAACL, ACL-IJCNLP. In addition, I review for conferences and workshops in extracting knowledge at scale: AKBC, Scholarly Document Processing Workshop, and others. I review for CoLM (2025, 2026). I have reviewed for NSF SBIR Phase I grants (2021, 2024). I have reviewed for the Cochrane Research Synthesis Methods Journal (2024, 2026) and Leibniz Collaboration (2026). While on the East Coast I occasionally volunteered with the Boston Appalachian Mountain Club chapter (2023-5).

INVITED TALKS (Excluding hiring talks)

MILA NLP Methods for Systematic Reviewing	Montreal, QC September, 2023
CoRise Guest Lecture ERASER: Evaluation of Interpretability Methods	CoRise February, 2023
Ought Building a Multi-Document Medical Summarization Dataset	San Francisco, CA July, 2022

PUBLICATIONS

In Submission

- [1] Anirudh Ajith, Amanpreet Singh, **Jay DeYoung**, Nadav Kunievisky, Austin C. Kozlowski, Oyvind Tafjord, James Evans, Daniel S. Weld, Tom Hope, and Doug Downey. “PreScience: A Dataset and Benchmark for Scientific Forecasting”. In: *ArXiv* abs/2602.20459 (2026).
- [2] Dany Haddad, Dan Bareket, Joseph Chee Chang, **Jay DeYoung**, Jena D. Hwang, Uri Katz, Mark Polak, Sangho Suh, Harshit Surana, Aryeh Tiktinsky, Shriya Atmakuri, Jonathan Bragg, Mike D’Arcy, Sergey Feldman, Amal Hassan-Ali, Rubén Lozano, Bodhisattwa Prasad Majumder, Charles McGrady, Amanpreet Singh, Brooke Vlahos, Yoav Goldberg, and Doug Downey. “Understanding Usage and Engagement in AI-Powered Scientific Research Tools: The Asta Interaction Dataset”. In: *ArXiv* abs/2602.23335 (2026).
- [3] Jungsoo Park, Hyungjoo Chae, Ethan Mendes, **Jay DeYoung**, Varsha Kishore, Wei Xu, and Alan Ritter. “Distribution-Aware Reward: Reinforcement Learning over Predictive Distributions for LLM Regression”. In: *ArXiv* abs/2605.20740 (2026).

Conference & Journal

- [4] Xinran Zhao, Aakanksha Naik, **Jay DeYoung**, Joseph Chee Chang, Jena D. Hwang, Tongshuang Wu, and Varsha Kishore. “Improving Attributed Long-form Question Answering with Intent Awareness”. In: *The Fourteenth International Conference on Learning Representations*. 2026.
- [5] Gaelen P. Adam, **Jay DeYoung**, Alice Paul, Ian J. Saldanha, Ethan M. Balk, Thomas A. Trikalinos, and Byron C. Wallace. “Literature search sandbox: a large language model that generates search queries for systematic reviews”. In: *JAMIA Open* 7.3 (2024), ooae098.
- [6] **Jay DeYoung**, Stephanie C. Martinez, Iain J. Marshall, and Byron C. Wallace. “Do Multi-Document Summarization Models Synthesize?” In: *Transactions of the Association for Computational Linguistics* 12 (2024), pp. 1043–1062.
- [7] Somin Wadhwa, **Jay DeYoung**, Benjamin Nye, Silvio Amir, and Byron Wallace. “Jointly Extracting Interventions, Outcomes, and Findings from RCT Reports with LLMs”. In: *Proceedings of the Machine Learning for Healthcare Conference*. New York, NY, USA, 2023.
- [8] Lucy Lu Wang, Yulia Otmakhova, **Jay DeYoung**, Thinh Hung Truong, Bailey Kuehl, Erin Bransom, and Byron Wallace. “Automated Metrics for Medical Multi-Document Summarization Disagree with Human Evaluations”. In: *ACL*. Toronto, CA, 2023.
- [9] **Jay DeYoung**, Han-Chin Shing, Luyang Kong, Christopher Winestock, and Chaitanya P. Shivade. “Entity Anchored ICD Coding”. In: *AMIA Annual Symposium* (2022).
- [10] **Jay DeYoung**, Iz Beltagy, Madeleine van Zuylen, Bailey Kuehl, and Lucy Lu Wang. “MS²: Multi-Document Summarization of Medical Studies”. In: *EMNLP*. 2021.

- [11] Benjamin E. Nye, **Jay DeYoung**, Eric P. Lehman, Ani Nenkova, Iain James Marshall, and Byron C. Wallace. “Understanding Clinical Trial Reports: Extracting Medical Entities and Their Relations”. In: *AMIA Summits on Translational Science Proceedings*. 2021.
- [12] **Jay DeYoung**, Sarthak Jain, Nazneen Fatema Rajani, Eric Lehman, Caiming Xiong, Richard Socher, and Byron C. Wallace. “ERASER: A Benchmark to Evaluate Rationalized NLP Models”. In: *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*. Online: Association for Computational Linguistics, July 2020, pp. 4443–4458.
- [13] Eric Lehman, **Jay DeYoung**, Regina Barzilay, and Byron C Wallace. “Inferring Which Medical Treatments Work from Reports of Clinical Trials”. In: *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics* (2019).
- [14] Rabih Zbib, Lingjun Zhao, Damianos G. Karakos, William Hartmann, **Jay DeYoung**, Zhongqiang Huang, Zhuolin Jiang, Noah Rivkin, Le Zhang, Richard M. Schwartz, and John Makhoul. “Neural-Network Lexical Translation for Cross-lingual IR from Text and Speech”. In: *Proceedings of the 42nd International ACM SIGIR Conference on Research and Development in Information Retrieval* (2019).
- [15] Ryan Gabbard, **Jay DeYoung**, Constantine Lignos, Marjorie Freedman, and Ralph Weischedel. “Combining rule-based and statistical mechanisms for low-resource named entity recognition”. In: *Machine Translation* 32.1-2 (2018), pp. 31–43.
- [16] Adrian Benton, **Jay DeYoung**, Adam Teichert, Mark Dredze, Benjamin Van Durme, Stephen Mayhew, and Max Thomas. “Faster (and better) entity linking with cascades”. In: *NIPS Workshop on Automated Knowledge Base Construction*. 2014.

Workshop

- [17] Lucy Lu Wang, **Jay DeYoung**, and Byron Wallace. “Overview of MSLR2022: A Shared Task on Multi-document Summarization for Literature Reviews”. In: *SDP*. 2022.
- [18] **Jay DeYoung**, Eric Lehman, Benjamin Nye, Iain Marshall, and Byron C. Wallace. “Evidence Inference 2.0: More Data, Better Models”. In: *Proceedings of the 19th SIGBioMed Workshop on Biomedical Language Processing*. Online: Association for Computational Linguistics, July 2020, pp. 123–132.
- [19] **Jay DeYoung**, Yee Seng Chan, Chinnu Pittapally, Hannah Provenza, Ryan Gabbard, and Marjorie Freedman. “BBNs 2017 KBP EAL System”. In: *TAC*. 2017.
- [20] Manaj Srivastava, David Trupiano, David Akodes, Ilana Heintz, Hannah Provenza, Bonan Min, **Jay DeYoung**, Lance A Ramshaw, and Roger Bock. “Adept Automatic Knowledge Discovery System for Cold Start Knowledge Base Population.” In: *TAC*. 2017.

Unpublished & Preprint

- [21] Ryan Gabbard, **Jay DeYoung**, and Marjorie Freedman. “Events Beyond ACE: Curated Training for Events”. In: *arXiv preprint arXiv:1809.05576* (2018).